

13.2 Self-Consistency and Tree-of-Thoughts

Sample several traces and vote. Search a tree of partial thoughts when one path is not enough.

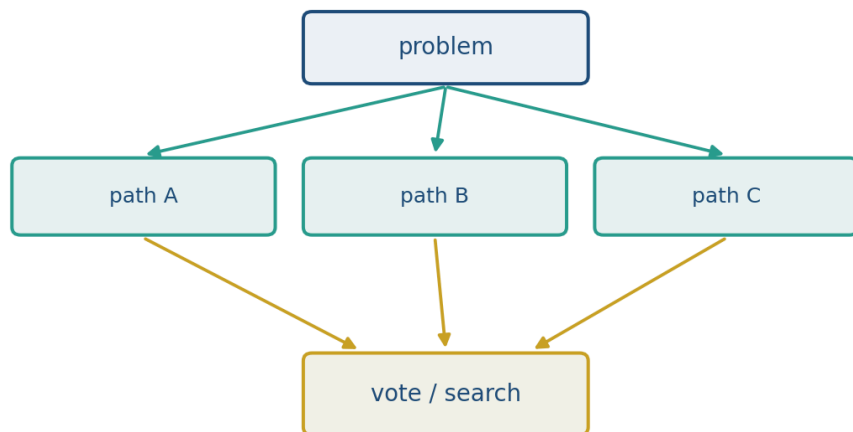
These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

One chain is a sample, not a proof. **Self-consistency** asks for several traces and **votes**. **Tree-of-thoughts** (ToT) searches among partial plans instead of committing to a single left-to-right story.

1. Majority vote over traces

Sample k independent CoT traces (temperature > 0). Parse an answer from each. Take the majority. The hope: wrong paths disagree; right paths land on the same number.

Self-consistency votes; Tree-of-Thoughts searches partial steps



This is test-time compute. You pay k forwards. Report k and the parser (last integer, boxed span, regex). Voting on the **answer**, not on the wording of the steps, is the usual rule. Lab 13 does that vote on constructed strings. Plot accuracy against k if you use this in a project; the curve usually flattens.

Self-consistency does not fix a shared bug. If every trace uses the same wrong formula, the majority is still wrong.

2. Search, not just i.i.d. samples

ToT treats a “thought” as a node: a subgoal, a candidate equation, a next move in a game. You expand a few children, **score** them (a prompt, a heuristic, or a checker), and keep a beam or run BFS/DFS. That is search with an LM as the proposal distribution. Greedy CoT is a tree of depth equal to one chain and width 1. ToT buys width.

ReAct (Week 14) is a different control loop: the node can be a **tool call**, not only more text. Do not mix the names in the report. Vote versus tree versus tools are three knobs.

3. What to measure

Accuracy versus k is the self-consistency plot. For ToT, accuracy versus expansion budget (nodes scored). Always include a greedy CoT baseline at $k = 1$. Cost belongs in the table: tokens, or wall time on the hardware you used.

4. Practice

1. Why does self-consistency need temperature greater than 0?
2. Give one failure that majority vote cannot catch.
3. ToT scores partial thoughts. Why is a cheap checker (arithmetic, a unit test) a better scorer than another unconstrained prompt, when you have one?